DDS 47–51 MHz | 1 kHz step | −10 dBm into 50 ohm
1508PL8T + 1887VE7T | +15 V | target PCB <=80 x 50 mm

REVISION B — ENGINEERING DRAFT
All circuit sections connected. G1 blocked pending manufacturer passport.
PCB placement, thermal/noise validation and output calibration remain open.
Matching global labels connect electrically across all sheets.

01_power

01_power.kicad_sch

02_dds

02_dds.kicad_sch

03_bias

03_bias.kicad_sch

04_bypass

04_bypass.kicad_sch

05_mcu

05_mcu.kicad_sch

06_ui

06_ui.kicad_sch

07_spi

07_spi.kicad_sch

08_clock

08_clock.kicad_sch

09_rf

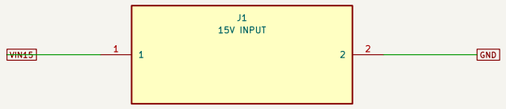
09_rf.kicad_sch

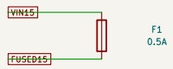
Read README_RU.md for assembly choices, firmware contract, sources and validation limits.
Open DDS_47_51.kicad_pro. PDF and netlist come from this same schematic.

Sheet: /
File: DDS_47_51.kicad_sch

Title: DDS 47–51 MHz / revision B / circuit draft

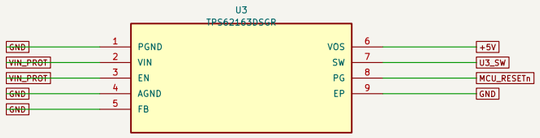
Size: A3	Date: 2026-10-29	**Rev: B / engineering draft**
KiCad E.D.A. 10.99.0-6047-ga25bc571ae		Id: 1/10







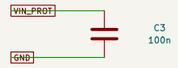
Input +15 V +/−5%; VIN must stay <17 V.
D1 blocks reverse polarity.

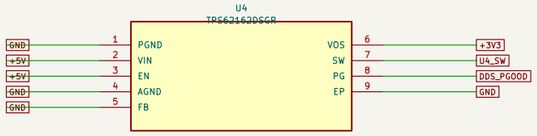








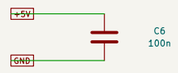


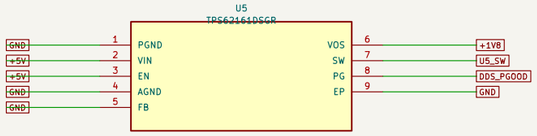


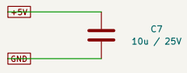






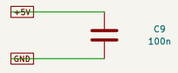


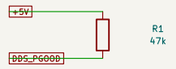












Capacitance specified EFFECTIVE at DC bias; X7R. EP=GND. Keep SW loops short.
U4/U5 PG wired-AND. MCU checks rails after startup delay; PG is Hi-Z when disabled.
Budget: 1.8 V <=0.4 A DDS; 3.3 V DDS <=0.2 A plus oscillator. Verify G1 current.

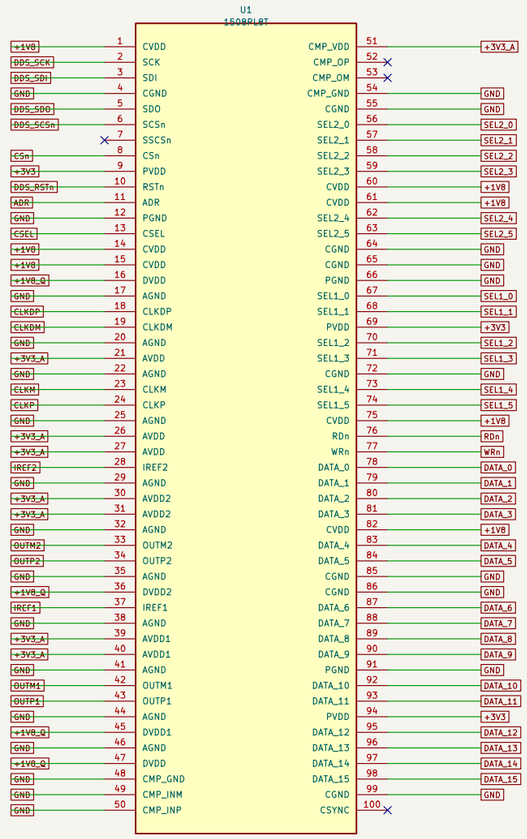
Sheet: /01_power/
File: 01_power.kicad_sch

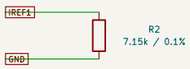
Title: Power / regulated 15 V input

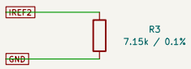
Size: A3	Date: 2026-10-29	**Rev: B / engineering draft**
KiCad E.D.A. 10.99.0-6047-ga25bc571ae		Id: 2/10

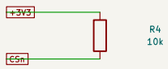
ELVEES / 14.10.2020
Table 4.2 pin numbering
CSEL=0: G1 / CLKP,M
CSEL=1: external input
480 MHz / CTR.cmx=0

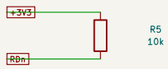
Firmware initialization:
DAC2 and comparator OFF
SYNC outputs OFF
LINK OFF / profile 0
Unused IO pulled down

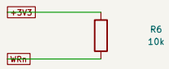


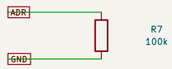










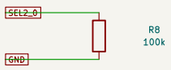


Continuous common ground plane. RREF: nominal IFS=10.07 mA. OUT voltage limit +/−0.6 V.

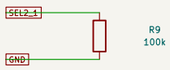
Sheet: /02_dds/
File: 02_dds.kicad_sch

Title: DDS / 1508PL8T / all 100 physical pins

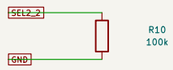
Size: A3 | Date: 2026-10-29 | **Rev: B / engineering draft**
KiCad E.D.A. 10.99.0-6047-ga25bc571ae | Id: 3/10



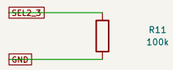
R8
100k



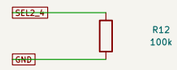
R9
100k



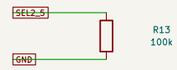
R10
100k



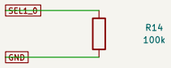
R11
100k



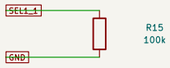
R12
100k



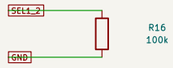
R13
100k



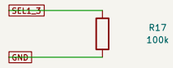
R14
100k



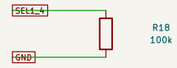
R15
100k



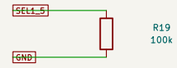
R16
100k



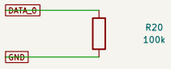
R17
100k



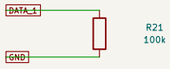
R18
100k



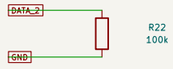
R19
100k



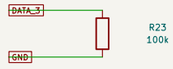
R20
100k



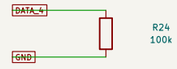
R21
100k



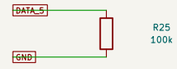
R22
100k



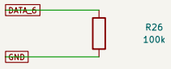
R23
100k



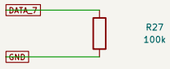
R24
100k



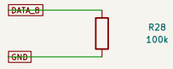
R25
100k



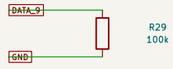
R26
100k



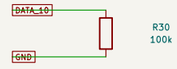
R27
100k



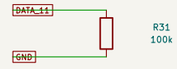
R28
100k



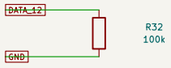
R29
100k



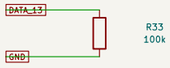
R30
100k



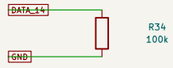
R31
100k



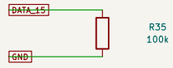
R32
100k



R33
100k



R34
100k



R35
100k



FB1
600R@100MHz / 1A



FB2
600R@100MHz / 1A



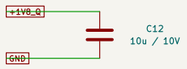
FB3
600R@100MHz / 1A



C10
10u / 10V



C11
10u / 10V



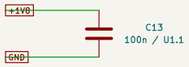
C12
10u / 10V

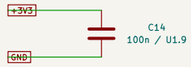
FB1..3 are ferrite beads, DCR <=0.1 ohm. Check LC resonance on prototype.
Unused bidirectional IO use resistive pull-downs, never direct ground shorts.

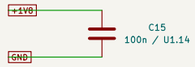
Sheet: /03_bias/
File: 03_bias.kicad_sch

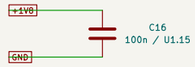
Title: DDS / unused inputs and supply filtering

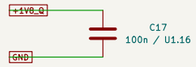
Size: A3	Date: 2026-10-29	**Rev: B / engineering draft**
KiCad E.D.A. 10.99.0-6047-ga25bc571ae		Id: 4/10

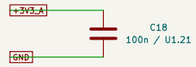


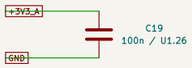


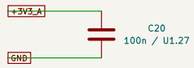


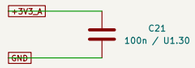








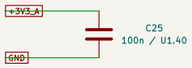


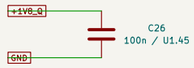


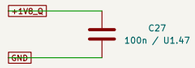






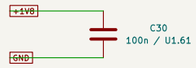


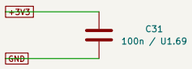








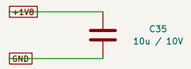


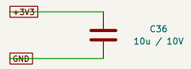


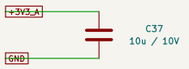


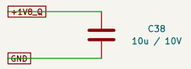










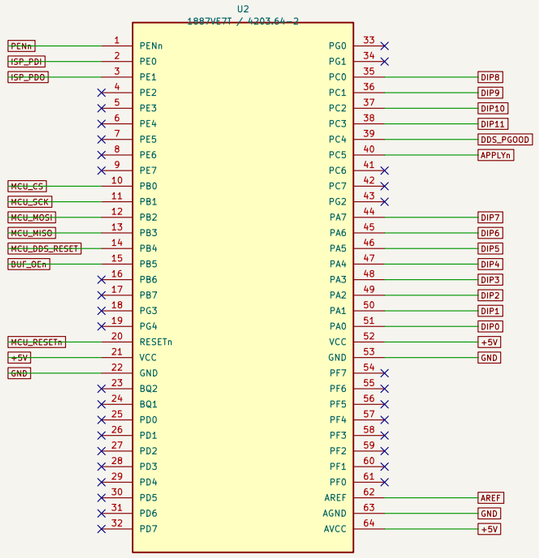


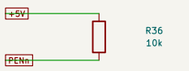
Place each capacitor at the indicated U1 supply pin with its own adjacent ground via.
Bulk capacitors supplement local bypass. Do not place all capacitors in a remote bank.

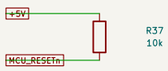
Sheet: /04_bypass/
File: 04_bypass.kicad_sch

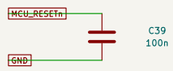
Title: DDS / capacitor at every supply pin

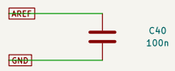
Size: A3	Date: 2026-10-29	**Rev: B / engineering draft**
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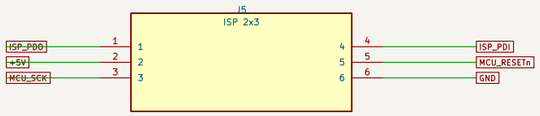


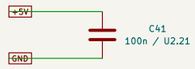


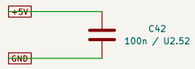












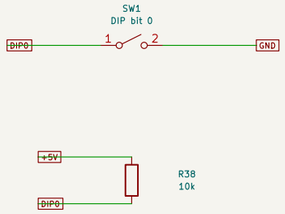


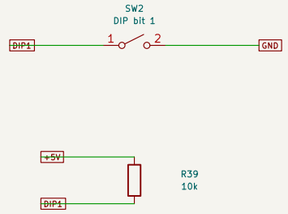
Configure internal RC 8 MHz. AVCC tied directly to VCC. Disable unused peripherals.
ISP uses PE0/PE1, not PB2/PB3. J5.2=target voltage sense; programmer must not power target.
NC GPIO: configure as inputs with pull-ups; BQ1/BQ2 unused in internal RC mode.

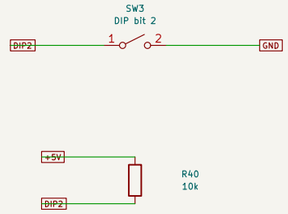
Sheet: /05_mcu/
File: 05_mcu.kicad_sch

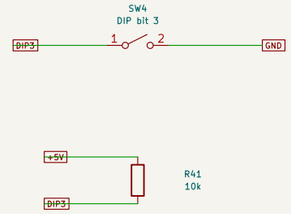
Title: Controller / 1887VE7T / internal RC / 5 V

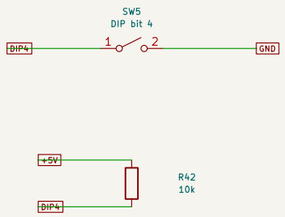
Size: A3	Date: 2026-10-29	**Rev: B / engineering draft**
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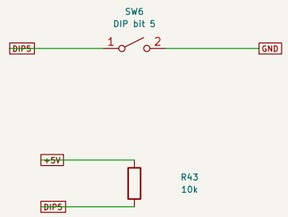


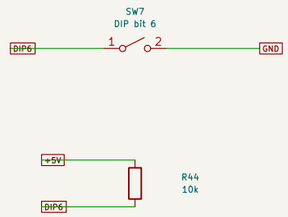


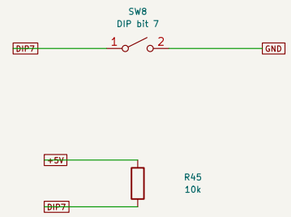


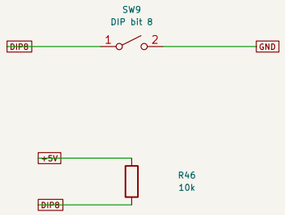


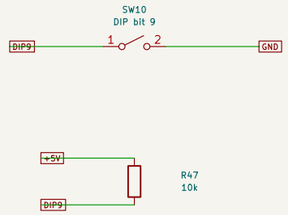


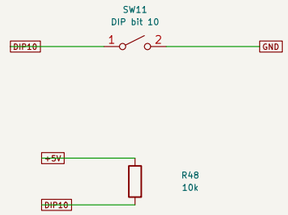


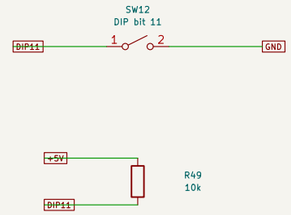


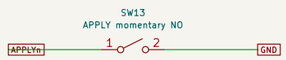




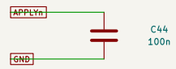










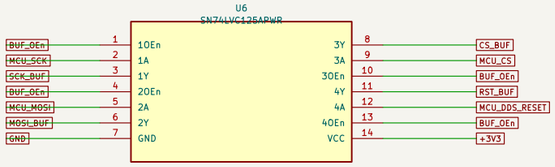


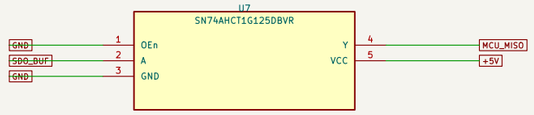
ON=bit 1: invert GPIO. N=sum(bit[i]*2^i). f=47,000,000+1,000*N Hz.
Accept N=0..4000; reject 4001..4095. Debounce APPLY and read stable DIP twice.
SW1..12 may be grouped into physical DIP packages when assigning footprints.

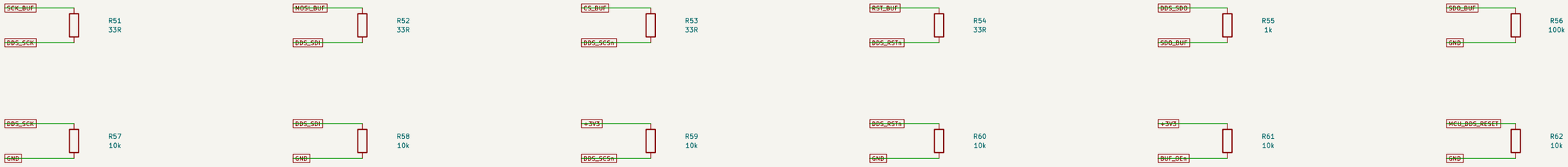
Sheet: /06_ui/
File: 06_ui.kicad_sch

Title: Frequency selector / 12 contacts and APPLY

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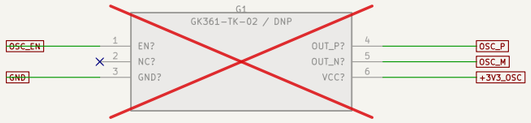


U6 disabled at startup. PB5 open-drain emulation: output LOW or input, never output HIGH.
Set CS=1/SCK=0/MOSI=0/RESET=0, wait rails and clock; enable U6, then release DDS reset.
Start SPI at 100 kHz / MSB first / 24-bit commands. Keep U6 disabled during ISP.

Sheet: /07_spi/
File: 07_spi.kicad_sch

Title: SPI / 5 V to 3.3 V translation

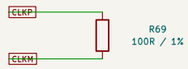
Size: A3	Date: 2026-10-29	**Rev: B / engineering draft**
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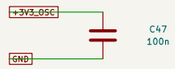


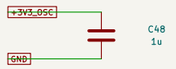


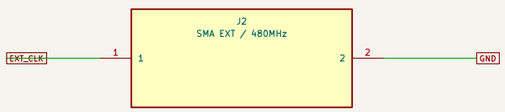


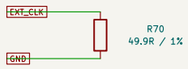






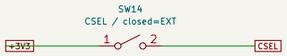


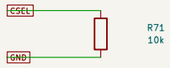












Default: G1 and 0R links DNP; SW14 closed. J2: 480 MHz sine / 0 dBm / 50 ohm.
G1 option: verify manufacturer pinout, 480 MHz grade, LVDS, enable and current BEFORE fitting.
Then fit G1/links and open SW14. G1 pair: 100-ohm differential. No guessed footprint.
U1 clock input has internal common-mode bias; external input is AC coupled.

Sheet: /08_clock/
File: 08_clock.kicad_sch

Title: 480 MHz clock / oscillator option and external input

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OUTP1
R72
49.9R / 0.1%
GND
OUTM1
R73
49.9R / 0.1%
GND
OUTP2
R74
49.9R / 0.1%
GND
OUTM2
R75
49.9R / 0.1%
GND
OUTP1
C51
100p C0G
L4
21nH
L5
200nH
L6
200nH
L7
21nH
R76
4.7R
RF_OUT
C52
56p C0G
GND
C53
91p C0G
GND
C54
56p C0G
GND
R77
866R 1%
GND
R78
866R 1%
GND
J3
SMA OUT / 50 ohm
RF_OUT
1
1
2
2
GND
OUTM1 has matching DC termination. IFS nominal 10.07 mA.
Unloaded pin swing <=0.503 V nominal (limit +/-0.6 V).
7-pole Butterworth fc~70 MHz; ideal values rounded to E24.
Pi attenuator ~1 dB. Set -10 dBm by calibrated DDS amplitude.
Use RF high-Q inductors; verify Q/SRF with vendor models.
No measured SFDR/noise or production compliance claimed.
Sheet: /09_rf/
File: 09_rf.kicad_sch
Title: RF / DAC loads / 7-pole LPF / pi pad / output
Size: A3
Date: 2026-10-29
Rev: B / engineering draft
KiCad E.D.A. 10.99.0-6047-ga25bc571ae
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